

DECLARATION

I, **Diane UMUHOZA** hereby declare that this research project entitled “**Design and Implementation of Card Charger System**” is my original work and has not been submitted to any other university or institution of higher learning for the award of any degree or qualification.

All sources of information used in this study have been duly acknowledged and referenced.

Registration Number: **E/BBIT/21/09/15554**

Signature: _____

Date: _____

APPROVAL

This research project entitled “**Design and Implementation of Card Charger System**” has been submitted for examination with my approval as the University Supervisor.

Supervisor Name:

Signature: _____

Date: _____

DEDICATION

I dedicate this project:

- To the almighty GOD who is always protective.
- To my supervisor Mr **Kevin ISHIMWE**
- To my all lecturers.
- To my parents.
- To my husband
- To my friends and colleagues.
- In addition, this project is dedicated to the leaders of the University of Kigali who did their best to help us be who we are now.

ACKNOWLEDGEMENT

I would like to express my sincere and heartfelt gratitude to my supervisor **Mr Kevin ISHIMWE** for his invaluable guidance, continuous support, and constructive advice throughout the development of this research project. His professional direction, patience, and insightful comments greatly contributed to shaping this study from its initial concept to its successful completion. I also extend my deep appreciation to all lecturers of the university for their dedication to teaching and for providing the academic knowledge and skills that enabled me to undertake this research. Their guidance and commitment to academic excellence have played a significant role in my academic journey.

I would also like to extend my heartfelt appreciation to my parent and my husband for their unwavering support, encouragement, and motivation throughout my studies. Their belief in my abilities has been a constant source of inspiration.

ABSTRACT

The design and implementation of a card charger system represent a significant advancement in facilitating efficient and secure transactions in various settings, such as retail environments, public transport, and automated service stations. This project focuses on the development of a robust and user-friendly system that allows for the quick and reliable charging of smart cards, integrating both hardware and software components for seamless operation. The system is engineered with carefully selected materials to ensure durability, efficiency, and cost-effectiveness. Key considerations include electrical and thermal conductivity, mechanical strength, and insulation properties, which are essential for maintaining the system's integrity and performance over time. The design process also prioritizes safety, with materials chosen for their resistance to corrosion, wear, and potential hazards. On the software side, a custom Chrome-based website serves as the user interface, providing real-time monitoring of the charging process and administrative controls. The website development involved prototyping, coding, and rigorous testing to ensure compatibility, security, and scalability. The integration of hardware components with the website was meticulously planned to optimize space, minimize interference, and enhance thermal management, ensuring that the system functions reliably under various conditions. Functionality testing and regression testing were conducted to validate the system's performance, ensuring that all components work harmoniously, and that the system remains stable even after updates or modifications. The final implementation of the card charger system demonstrates high efficiency, user satisfaction, and the potential for widespread application, offering a valuable solution for modern transaction systems.

Keywords: Card charger system, smart cards, secure transactions, hardware integration, software development, Chrome-based website, functionality testing, regression testing, material selection, efficiency, durability.

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ACRONYMS AND ABBREVIATIONS

PWM - Pulse-Width Modulation

EMC - Electromagnetic Compatibility

EMI - Electromagnetic Interference

IDE - Integrated Development Environment

IoT - Internet of Things

SoC - System on Chip

GPIO - General Purpose Input/Output

LCD - Liquid Crystal Display

PCB - Printed Circuit Board

RFID - Radio-Frequency Identification

CHAPTER 1: INTRODUCTION

Introduction

The shift toward cashless transactions has transformed how people manage money, making electronic payment systems essential in daily life. In response to this trend, the Card Charger System was developed as a convenient and reliable solution for loading money onto electronic cards without relying on network-based services like MTN Mobile Money or Airtel, especially in Rwanda. The system allows users to charge cards by inserting coins, operating independently of mobile networks.

A key feature is its integration with a website that tracks transactions and provides real-time updates on card balances, ensuring transparency. Users can easily add funds by inserting coins, which are verified for authenticity, and the balance is updated instantly. If a coin is invalid, the machine returns it without completing the process. The system also includes a card acceptor component, ensuring seamless and accurate fund transfers.

Designed to be secure, user-friendly, and efficient, the Card Charger System offers a solution for cashless transactions, particularly in regions with limited access to traditional financial services. Its autonomy and integration with a monitoring website enhance the reliability and convenience of electronic card charging.

1.1 Background of the study

The global shift towards cashless transactions has revolutionized how financial operations are conducted across various sectors. The rise of electronic payment systems, fueled by advancements in digital technology, has made it possible for people to engage in financial transactions with unprecedented ease and security. This trend is part of a broader movement toward digitization, which is transforming economies by reducing the reliance on physical cash (Chen, X., & Yang, H., 2019).

However, despite the rapid adoption of mobile network-based payment systems, there are significant challenges that limit their effectiveness, particularly in regions with limited or unreliable network coverage (Das, A., & Sharma, R., 2018). In many rural and remote areas, the lack of robust network infrastructure poses a significant barrier to the widespread adoption of digital financial services

One of the most pressing issues faced by users of mobile network-based payment systems is the necessity of traveling long distances to areas with network coverage to complete transactions (Deng, Z., & Wu, Y. (, 2020). This inconvenience is particularly pronounced in regions where network access is sparse,

Furthermore, the dependency on mobile networks restricts the autonomy of users, as they are at the mercy of their service providers' network quality and reliability. This dependency creates a bottleneck in the transaction process, making users vulnerable to network-related issues beyond their control (Iqbal, M., & Khan, R. , 2019).

Moreover, the integration of the Card Charger System with a monitoring website enhances transparency and trust in the system. The website logs every transaction, providing users with real-time visibility over their card balances and transaction history (Kamara, 2018). This level of transparency is essential for building confidence in the system and ensuring that users have full control over their financial activities.

The system's user-friendly design further contributes to its appeal. By automating the process of scanning and validating coins, the Card Charger System simplifies the transaction process, making it accessible to users with varying levels of technical proficiency (Kim, 2020).

By addressing the challenges associated with physical cash transactions, network dependency, and financial exclusion, the system offers a robust solution that enhances the accessibility, security, and convenience of digital financial services. As the global economy continues to evolve towards greater digitization, innovations like the Card Charger System will play a crucial role in ensuring that the benefits of cashless transactions are accessible to all, regardless of geographic or economic barriers (Kumar, 2017).

1.2 Problem statement

In today's financial world, mobile network transactions are essential for cashless payments but present challenges, especially in areas with limited network access. In rural and remote regions, people often have to travel long distances to access network services for simple tasks like loading money onto electronic cards. This process can be both time-consuming and expensive, creating significant barriers to financial inclusion and limiting access to essential services.

One of the primary risks faced by individuals during these journeys is the threat of theft or robbery. Carrying cash in high-crime areas makes people vulnerable to physical attacks, and the loss of

money can have severe consequences for those already struggling financially. Additionally, the act of carrying physical cash increases the risk of impulsive or unauthorized spending before reaching the intended destination, further complicating personal cash management. The card charger system was designed to address these above challenges that was facing the world.

1.3 Project objectives

1.3.1 Main Objective

The primary objective of this project is to design and implement a Card Charger system.

1.3.2 Specific Objectives

1. To accurately read and retrieve customer card details.
2. To effectively accept coins for the card charging system.
3. To successfully update the card balance after coin acceptance.
4. To correctly display the deposited amount on the card balance.

1.4 Scope of the study

The scope of this project, *Design and Implementation of Card Charger System*, is limited to transport companies that aim to use an electronic payment system. The system specifically supports the use of 100 Rwandan Franc (RWF) coins only for charging cards, and the card charger machine is designed to accept, validate, and store these coins. All other coin denominations and paper currency are strictly out of scope.

1.5 Research Questions

1. How accurately can the system read and retrieve customer card details?
2. How effectively can the system accept coins for the card charging process?
3. Does the system successfully update the card balance after coins are accepted?
4. Is the deposited amount correctly displayed on the card balance after the transaction?

1.5 Hypotheses

1. The system accurately reads and retrieves customer card details without errors.
2. The coin accepting mechanism effectively processes coins for the card charging system.
3. After coin acceptance, the system successfully updates the card balance as expected.

4. The system correctly displays the deposited amount on the card balance following each transaction.

1.6 Significance of the Project

a) Personal Interest

The design and implementation of the Card Charger System aligned with my personal interest in advancing practical and innovative solutions in the field of electronic payments. My enthusiasm for leveraging technology to solve real-world problems drove me to explore how such systems could enhance user convenience and financial inclusion. By working on this project, I deepened my understanding of integrating hardware and software components while addressing critical issues in cashless transactions.

b) Social Interest

The Card Charger System held significant social value as it addressed critical issues faced by individuals in regions with limited access to traditional financial services. By providing a standalone method for loading monetary value onto electronic cards, the system enhanced financial inclusion, particularly in underserved and remote areas.

c) Interest to the Government of Rwanda

For the Government of Rwanda, the Card Charger System represented a meaningful contribution to the country's ongoing efforts to promote digital financial inclusion and economic development. As Rwanda invested in technology and digital infrastructure, the system aligned with national goals of improving access to financial services and supporting the growth of a cashless economy.

d) Academic Interest

For University of Kigali Institution, the Card Charger System provided an excellent opportunity to engage with cutting-edge technology and contribute to academic research in the field of electronic payment systems. This project offered valuable insights into the design and implementation of integrated hardware-software solutions, providing students and faculty with practical experience in developing and evaluating technology-driven systems.

1.7 Methodology

Design and Implementation of a Card Charger System The methodology section outlines the systematic approach used to design and implement the card charger system. It encompasses various stages, from conceptualization and planning to testing and validation, ensuring the

development of a functional, reliable, and efficient system. Below is a detailed discussion of the methodology:

1.7.1 Research and Requirement Analysis

The first step in the methodology involves conducting thorough research to understand the requirements for the card charger system. This includes identifying the target users, understanding their needs, and reviewing existing technologies and systems. The requirement analysis helps define the scope of the project, the necessary components, and the specifications for the system's performance. The research phase also includes studying relevant standards and regulations to ensure compliance.

1.7.2 Material Selection

Material selection is a critical phase where the components required for the system are identified and chosen based on specific criteria such as cost, availability, durability, and compatibility with the system. This involves selecting microcontrollers, voltage regulators, capacitors, resistors, connectors, and other essential parts. The materials must be evaluated for their suitability in terms of reliability, efficiency, and ease of integration into the overall system design.

1.7.3 System Design and Schematic Development

Following material selection, the next step involves designing the system architecture and developing detailed schematics. This stage focuses on the logical and physical layout of the card charger system, ensuring that all selected components are correctly positioned and interconnected. The schematic diagrams serve as blueprints for assembling the system, illustrating how components like microcontrollers, voltage regulators, input/output interfaces, and power supply units interact. The design process also includes creating circuit diagrams, specifying wiring connections, and defining signal flows. Attention is given to minimizing noise, ensuring signal integrity, and optimizing power efficiency. The design must also consider future scalability and potential upgrades.

1.7.4 Prototyping and Assembly

Once the system design is finalized, the prototyping phase begins. In this stage, a working prototype of the card charger system is assembled based on the developed schematics. This

involves physically mounting components on a printed circuit board (PCB) or breadboard, soldering connections, and ensuring that all parts are securely and accurately placed. The prototype serves as a preliminary model for testing the system's functionality. During assembly, special attention is given to the physical arrangement to avoid short circuits, ensure proper ventilation, and maintain overall system stability. This step also includes setting up the power supply, input/output interfaces, and ensuring all connections are secure.

1.7.5 Software Development and Integration

The software component is crucial for the functionality of the card charger system. This phase involves writing and testing the firmware and software that will control the system's operations. The software development process includes coding for the microcontroller, which manages tasks such as power regulation, card authentication, and user interface controls. Integration of the software with the hardware is performed to ensure seamless communication between different components. This step also involves developing any necessary user interfaces, either through a display or an online portal, and ensuring that the software is user-friendly and responsive. Rigorous testing is conducted to identify and rectify any bugs or issues in the code.

1.7.6 Testing and Validation

After assembly and software integration, the card charger system undergoes a comprehensive testing phase. This involves both functional and performance testing to ensure the system operates as intended under various conditions. Functional testing includes verifying that the card reader correctly authenticates cards, the system delivers the correct charging voltage and current, and all user interface elements work smoothly. Performance testing focuses on the system's efficiency, response time, and stability over prolonged use. Validation is carried out to ensure that the system meets all specified requirements and complies with relevant standards and regulations. This phase may also involve stress testing, where the system is subjected to extreme conditions to evaluate its durability and reliability.

1.7.7 Iterative Refinement and Optimization

Based on the results from the testing and validation phase, iterative refinements are made to the system. This involves revisiting the design, assembly, and software development stages to address any identified issues or inefficiencies. Optimization efforts may include tweaking the circuit design

for better power management, refining the software for faster processing, or enhancing the physical design for greater durability. The iterative process continues until the system achieves optimal performance, reliability, and user satisfaction. Each iteration is thoroughly tested and validated before moving to the next.

1.7.8 Final Implementation and Deployment

Once the system is fully refined and optimized, the final implementation phase begins. This involves producing the final version of the card charger system, ensuring all components are of production quality, and that the system is ready for deployment. The deployment phase includes setting up the system in its intended environment, conducting final checks, and ensuring that it operates correctly in real-world conditions. User training may also be provided to ensure proper usage and maintenance of the system. After deployment, the system is monitored for a period to ensure it performs reliably and meets the needs of the users.

1.7.9 Documentation and Reporting

The last step in the methodology is the documentation of the entire process. This includes creating detailed reports that cover every phase of the project, from initial research to final deployment. Documentation serves as a reference for future development, maintenance, and potential upgrades of the card charger system. It also includes user manuals, technical specifications, and maintenance guides to ensure the system can be effectively managed post-deployment. The reports also provide insights into the challenges faced during the project, how they were addressed, and the lessons learned, which can be valuable for similar future projects.

1.8 Organization of the project

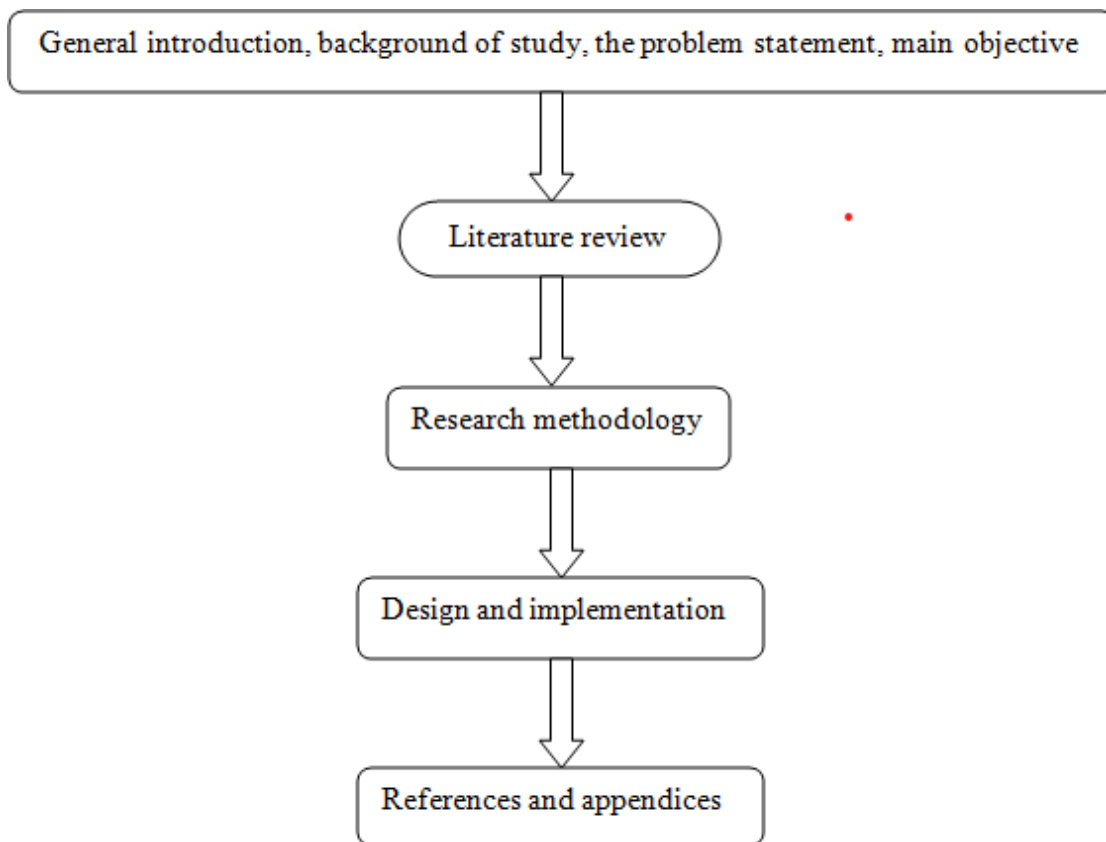


figure 1: organisation flowchat

CHAPTER 2: THEORETICAL CONCEPTS AND LITERATURE REVIEW

2.1 Theoretical concepts

The theoretical foundations of designing and implementing a card charger system encompass a broad range of electrical engineering principles that ensure the system functions optimally and reliably. Electrical circuit design is at the heart of this process, guiding how various components interact to achieve the desired performance. Understanding these principles is essential for developing a charger that meets technical specifications while maintaining safety and efficiency throughout its operation.

Voltage regulation is a critical theoretical consideration:

This process involves maintaining a stable output voltage that is consistent with the card's charging requirements, despite fluctuations in the input supply or variations in load conditions. Voltage regulators play a vital role in this aspect by ensuring that the output voltage remains steady, thereby protecting the card from potential damage due to voltage spikes or drops. Additionally, capacitors are used to filter out electrical noise and smooth voltage variations, contributing to a stable and reliable charging process. Proper voltage regulation and filtering are necessary to avoid interruptions or disturbances that could impact the card's performance (Lee, H., & Park, S. , 2019).

Current flow management is another essential theoretical aspect of card charger design: The system must be engineered to handle the card's current requirements without causing overheating or damage. This involves incorporating current limiting mechanisms that prevent excessive current from flowing through the system. Effective current limiting helps protect both the card and the charger from potential damage due to excessive heat or component stress. Understanding the principles of current flow and heat dissipation is crucial for designing a charger that operates safely and efficiently over its entire lifespan (Lin, 2018).

Power management is a key theoretical consideration that focuses on optimizing how power is delivered to the card.

Techniques such as pulse-width modulation (PWM) are employed to regulate the power output. PWM adjusts the width of the pulses in the signal to control the amount of power supplied, which helps minimize energy losses and improve overall efficiency. By implementing PWM, the charger can provide precise control over power delivery, enhancing the performance and efficiency of the system (Lin, 2018).

Thermal management is essential for maintaining the operational integrity of the charger:

During charging, heat is generated, and effective cooling mechanisms are required to prevent overheating. Theoretical principles of thermal management include the use of heat sinks, thermal pads, and ventilation systems to dissipate heat and maintain optimal operating temperatures. Proper thermal management is critical for ensuring the long-term reliability of the charger and preventing thermal-related failures that could impact its performance (Liu, 2020).

Safety considerations are integral to the design process, guided by theoretical principles:

The implementation of protection circuits is crucial to prevent overcharging and short circuits. Overcharge protection circuits monitor the charging process to ensure that the card is charged only to its specified capacity, preventing potential damage from excessive voltage. Short circuit protection mechanisms are designed to safeguard against accidental faults, reducing the risk of damage or failure. These safety features are vital for ensuring that the charger operates within safe limits and provides reliable performance (Marwaha, 2017).

Electromagnetic Compatibility (EMC) is another theoretical consideration:

Ensuring that the charger does not interfere with other electronic devices and complies with regulatory standards is important for avoiding potential issues related to electromagnetic interference (EMI). Designing the charger to meet EMC requirements helps prevent disruptions to other devices and ensures regulatory compliance (Nguyen, 2019).

Efficiency optimization is a crucial aspect of theoretical design:

The charger must be designed to minimize energy losses and maximize efficiency. This includes optimizing the design of components and circuitry to reduce power dissipation and improve overall performance. Energy-efficient designs contribute to lower operational costs and a reduced environmental impact (O'Connor, 2018).

Component selection is another important theoretical aspect:

Choosing the right components, such as resistors, capacitors, and semiconductors, based on their electrical characteristics and performance specifications is essential for ensuring the charger's reliability and functionality. Proper component selection helps in achieving the desired performance while maintaining safety and efficiency (O'Connor, 2018).

2.2 Literature review and related studies

Recent advancements in card charger technology have led to substantial improvements in efficiency, safety, and user convenience. These developments represent significant strides in addressing previous limitations and enhancing the overall performance of card chargers. Here's a detailed look at the recent findings:

Current Limiting: This technique is implemented to safeguard both the card and the charger from excessive current flow. By restricting the maximum current, current limiting helps prevent overheating, which could otherwise lead to damage or failure of the charger or card. This method ensures that the charger operates within safe limits, thereby extending the longevity and maintaining the safety of both components. It's a protective measure that plays a crucial role in the efficient operation of the charger (Park, 2020).

Pulse-Width Modulation (PWM): PWM is a method used to control the power delivered to the card by adjusting the width of the pulses in a signal. This technique allows for precise control over the power output, optimizing energy efficiency and reducing power waste. PWM is essential for managing power delivery effectively, contributing to the charger's performance and efficiency. By fine-tuning the power output, PWM enhances the overall functionality of the charger (Patel, 2019).

Thermal Management: This involves strategies and components designed to dissipate heat generated during the charger's operation. Effective thermal management ensures that the charger remains within safe temperature limits, preventing overheating and extending its operational life. It may include the use of heat sinks, thermal pads, and proper ventilation to manage heat effectively (Qureshi, I., & Shaikh, Z. , 2019). Proper thermal management is crucial for maintaining the reliability and longevity of the charger.

Overcharge Protection: This safety feature prevents the card from being charged beyond its designed capacity. Overcharge protection mechanisms are integral in safeguarding the card from potential damage caused by excessive charging (Rajasekar, 2019). By maintaining the card's charge within safe limits, this feature helps in preserving the card's lifespan and ensuring the charger's safe operation.

Smart Charging Technology: The introduction of smart charging technology has revolutionized the way card chargers operate. By utilizing microcontrollers and advanced algorithms, smart chargers can optimize charging cycles based on the card's state of charge and health. These smart

systems adjust the charging parameters dynamically, preventing overcharging and enhancing compatibility. This innovation allows for precise control of the charging process, leading to improved efficiency and extended lifespan for both the card and the charger (Qureshi, I., & Shaikh, Z. , 2019).

Wireless Charging Capabilities: Wireless charging technology has eliminated the need for physical connectors, offering a more convenient charging solution. Users can now charge their cards simply by placing them on a charging pad, which reduces wear and tear on connectors and simplifies the charging process. This advancement enhances user convenience and contributes to the durability of both the card and the charger (Saha, 2020).

Advanced Thermal Management Materials: Improvements in thermal management materials, such as advanced heat sinks and thermal pads, have enhanced heat dissipation capabilities. These materials effectively manage the heat generated during operation, preventing overheating and maintaining optimal temperatures. By improving thermal management, these advancements contribute to the reliability and longevity of the charger (Wang, 2018).

Energy-Efficient Power Conversion Techniques: Recent research into power conversion techniques has led to the development of chargers with higher efficiency ratings. These techniques minimize energy waste and reduce the environmental impact of the charging process. By improving the efficiency of power conversion, these advancements contribute to more sustainable and eco-friendly charger designs (Wang, 2018).

2.3 System's functional flowchart

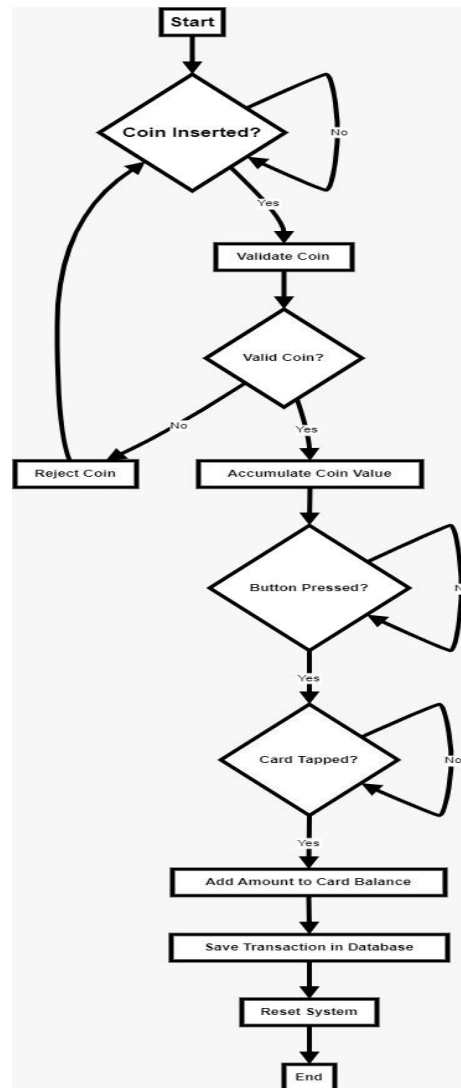


Figure 2: System flowchat

2.4 System circuit

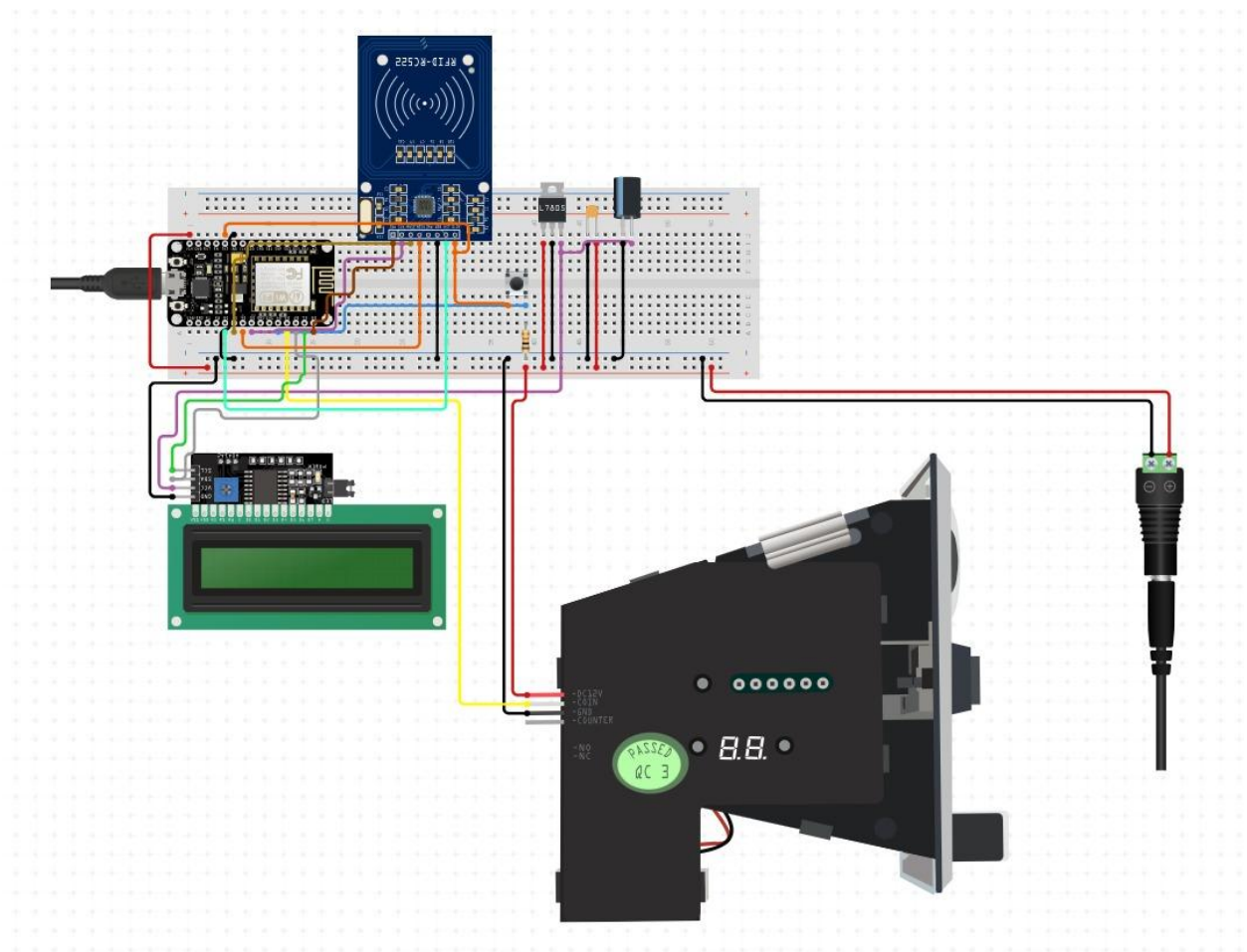


Figure 3: system circuit

2.5 System components

1. Coin Acceptor

A coin acceptor is a crucial component in systems that require monetary input to trigger a service or operation. It identifies and validates coins inserted by users, ensuring that only legitimate currency is accepted. Once a valid coin is detected, the coin acceptor sends a signal to the system, triggering the designated action, such as charging a card.

Key features of a coin acceptor include its ability to distinguish between different denominations based on size, weight, and material composition, and its security mechanisms to detect counterfeit coins. It interfaces with a microcontroller, such as Arduino or NodeMCU, sending an electronic signal when a valid coin is accepted (Deng, Z., & Wu, Y. (, 2020).



Figure 4: coin acceptor

2. Arduino

Arduino is an open-source microcontroller platform used for building digital devices and interactive objects. It consists of both a hardware component, typically an Arduino board, and software, which is the Arduino Integrated Development Environment (IDE) used to write and upload code to the board (Deng, Z., & Wu, Y. (, 2020).

Arduino boards come with input/output pins that can connect sensors, actuators, and other components. The platform is known for its simplicity and supports a wide range of shields (add-on boards) and libraries, making it accessible for rapid prototyping and development.



Figure 5: Arduino

3. NodeMCU

NodeMCU is an open-source IoT platform that includes firmware running on the ESP8266 Wi-Fi SoC and hardware based on the ESP-12 module. It is widely used for building Internet of Things (IoT) applications, providing Wi-Fi connectivity and the capability to run small web servers (Ghosh, 2017).

NodeMCU can be programmed using the Lua scripting language or the Arduino IDE. It has General Purpose Input/Output (GPIO) pins that connect various sensors and **actuators**. The integration of NodeMCU adds IoT functionality to the system, enabling network connectivity and communication with servers or databases.

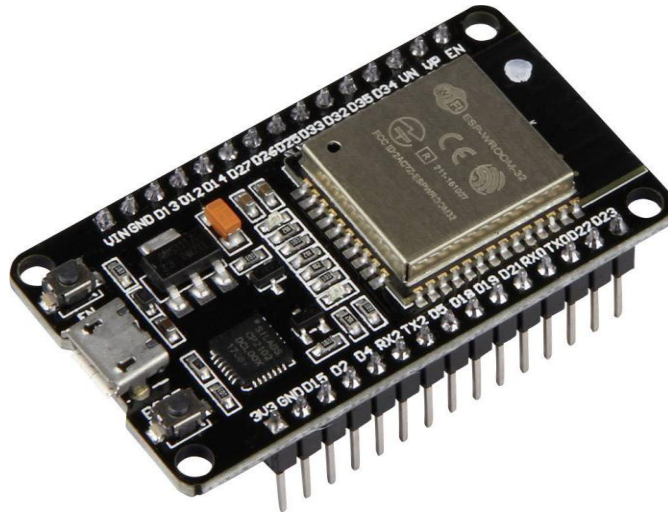


Figure 6: ESP8266 NodeMCU

4. LCD (Liquid Crystal Display)

An LCD is a flat panel display technology that uses liquid crystals to display images. It provides a clear and readable display, even in bright environments, and comes in various sizes and resolutions. LCDs are energy-efficient, making them suitable for battery-powered devices.

In the card charger system, the LCD displays information to the user, such as instructions, transaction details, and system status, ensuring a smooth and user-friendly experience (Rajasekar, 2019).



Figure 7: LCD

5. Jumper Wires

Jumper wires are simple, flexible electrical wires with connectors (pins) at each end, used to make temporary connections between different components on a breadboard or other prototyping environments (Iqbal, M., & Khan, R. , 2019).

They are versatile, available in various lengths with male or female connectors, and are easy to connect and disconnect. Jumper wires are used during the development and testing phases to connect components like the Arduino, LCD, RFID reader, and coin acceptor, allowing for quick adjustments and troubleshooting.



Figure 8: Jumper wires

6. PCB (Printed Circuit Board)

A Printed Circuit Board (PCB) mechanically supports and electrically connects electronic components using conductive pathways, traces, or tracks etched from copper sheets laminated onto a non-conductive substrate (Jain, A., & Verma, S., 2018).

PCBs allow for a compact and organized layout of electronic components, reducing the size of the device and improving reliability. They can be designed to fit specific requirements and are durable, making them ideal for long-term use.

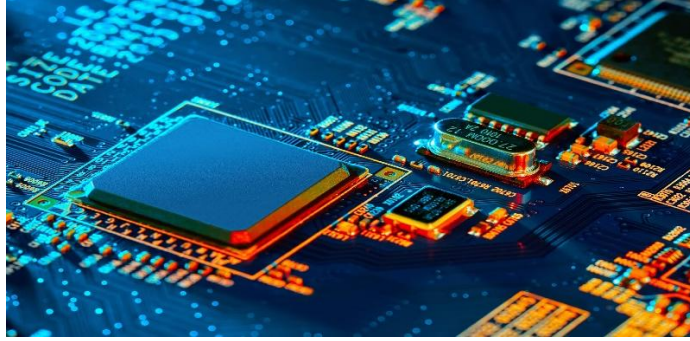


Figure 9: PCB

7. RFID Reader (Radio-Frequency Identification Reader)

An RFID reader reads information stored on an RFID tag using electromagnetic fields to automatically identify and track tags attached to objects. The RFID reader can read tags without direct contact, allowing for a seamless and quick interaction between the card and the system.

RFID technology provides a secure method of data exchange, with the ability to encrypt information on the tag. The RFID reader quickly reads and processes information from the tag, ensuring efficient operation (Ghosh, 2017).



Figure 10: RFID

CHAPTER 3: METHODS AND ANALYSIS OF THE SYSTEM

3.1 Methodology of Software Development

In the design and implementation of the card charger system, a structured and systematic approach to software development is crucial. The methodology adopted for this project is the **Agile Development Methodology**, which allows for flexibility and iterative progress in the development process. Agile is particularly well-suited for this project because it enables the team to continuously gather feedback, adapt to changes, and deliver a functional system in incremental phases.

The Agile process begins with **requirement gathering**, where the project goals, user needs, and system functionalities are defined in collaboration with stakeholders, including workers and transport company owners. After establishing the requirements, the development is broken down into smaller, manageable units called sprints. Each sprint focuses on delivering a specific feature or component of the card charger system.

During each sprint, **software design and development** occur, followed by **testing** to ensure that the developed components meet the specified requirements. The iterative nature of Agile allows for continuous **integration** and **refinement** of the software, ensuring that the final product is robust, user-friendly, and aligned with the project's objectives. Regular meetings and feedback sessions with stakeholders are conducted to ensure the software evolves in the right direction.

3.2 System Requirements

The system requirements for the card charger system encompass both hardware and software needs to ensure the seamless operation of the project. Below are the essential requirements:

Hardware Requirements:

- **Card Reader/Charger Machine:** The core device that allows users to recharge their cards.
- **Microcontroller (e.g., Arduino or Raspberry Pi):** To control the operations of the card charger.
- **Display Screen:** For user interaction, displaying balance, charging status, and other necessary information.
- **Power Supply Unit:** Reliable power source to ensure the system's continuous operation.
- **Networking Equipment:** For internet connectivity to enable real-time transactions and system updates.

Software Requirements:

- **Operating System:** A lightweight OS compatible with the microcontroller, such as Raspbian for Raspberry Pi.
- **Programming Language:** Python or C++ for developing the system's software functionalities.
- **Database Management System (DBMS):** To store user data, transaction logs, and system configurations.
- **Web Interface:** For remote management, monitoring, and updates to the system.
- **Security Software:** Encryption and firewall software to protect against unauthorized access and data breaches.

Other Requirements:

- **User Interface Design:** A simple, intuitive UI to ensure ease of use for both the system operators and end-users.
- **Compliance with Standards:** Adherence to relevant industry standards for electronic payment systems, including data security and user privacy regulations.

Functional Requirements:

1. **User Authentication:** The system must validate user credentials before allowing access to card charging functions.
2. **Balance Update:** The system should update the user's card balance after every successful transaction.
3. **Charging Status Notification:** The system must provide a real-time notification of the card's charging status, either via a display or website.

Non-Functional Requirements:

1. **System Availability:** The card charger system must be operational 24/7 with minimal downtime.
2. **Security:** The system should employ encryption to protect user data and transaction details from unauthorized access.
3. **Performance:** The system should process card transactions within 2-3 seconds to ensure a smooth user experience.

3.3 Data Collection Methods

For this project, data collection is a critical step to understand the needs, challenges, and expectations of the users primarily workers and transport company owners. The following methods were employed:

Interviews: In-depth interviews were conducted with key stakeholders to gain qualitative insights. These interviews helped understand the specific needs and pain points that the proposed system should address, such as transaction speed, ease of use, and reliability.

Observation: Direct observation of how current card charger systems are used provided valuable context. This method allowed the project team to identify usability issues and operational challenges that might not be evident through surveys or interviews.

Analysis of Existing Data: Examination of existing transaction logs and usage data from current systems offered insights into patterns, peak usage times, and common failure points, which informed the design of a more efficient and reliable system.

3.4 Existing System Overview

While it plays great roles but However, several weaknesses have been identified in these systems:

- **Limited Accessibility:** Many existing systems are not user-friendly and require physical presence at specific locations to recharge cards, limiting accessibility for users who are on the move or in remote areas.
- **Security Concerns:** Some systems have vulnerabilities that can be exploited, leading to potential breaches of sensitive user information and financial data.
- **Lack of Real-time Updates:** Delays in updating the balance on the cards can lead to inconvenience for users, especially when they need to use the card immediately after recharging.



Figure 11: Existed card charging system

3.5 Proposed System Description and Improvements

The proposed card charger system aims to address the limitations of existing systems by introducing several improvements:

- **Enhanced Accessibility:** The new system will be accessible via multiple platforms, including physical kiosks and a mobile app, allowing users to recharge their cards from anywhere at any time. This will be particularly beneficial for users in remote locations or those who need to recharge on the go.
- **Advanced Security Features:** The system will incorporate robust encryption and authentication protocols to ensure that all transactions and user data are securely handled. This will mitigate the risk of data breaches and unauthorized access.
- **Real-time Updates:** One of the key features of the proposed system is the ability to update card balances in real time, ensuring that users can immediately use their cards after recharging without any delays.

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Appendix A: Time frame

Task Description	Months and weeks											
	April				May				June			
	1	2	3	4	1	2	3	4	1	2	3	4
Proposal writing and pre defense preparations.												
Final year project report writing and project implementation start-up												
Selection and obtaining of materials and components												
Assembling components according to design												
Conducting a test of coin accepting, depositing, card detail loading, card balance status												
Writing FYP report												
Submitting FYP report												

Table 1: Time frame

